

MBA Fastrack 2025 (CAT + OMETs)

QUANTITATIVE APTITUDE

DPP: 8

Time Speed Distance II

- Q1** Riya and Sohan start simultaneously from A towards B and walk at speeds of 4 km/h and 10 km/h, respectively. At the same time, Priya starts walking from B towards A. Riya meets Priya exactly 5 minutes after Sohan meets Priya. If the distance between A and B is 3 km, find Priya's walking speed in km/h.
- Q2** Two runners are 338 meters apart and start running toward each other simultaneously. The first runner begins at a speed of 20 m/min, while the second starts at 30 m/min. However, both runners decrease their respective speeds by 2 m/min every minute. After how many minutes will they meet?
- Q3** Prateek started from point A at a speed of 24 km/hr towards point B which is 120 km away. After 20 minutes, Qutub started from point B at the speed of 36 km/hr towards point A. What is the distance between the point they will meet at and the point they would have met at had both of them started simultaneously?
- (A) 4.2 km (B) 4.8 km
(C) 5.4 km (D) 5.6 km
- Q4** A train 1250 meter long crosses a platform whose length is 950 meters in 40 seconds. If a dog is running at a speed of 3 m/s in the same direction as the train and the train is 208 meters behind the dog, then in how much time (in seconds) will the train catch the dog?
- Q5** Train A running at the speed of 90 km/hr crosses Ranjay who is running in the opposite direction at the speed of 18 km/hr in 24 sec. The length of train B is 140 m more than that of train A and is running at the speed of 126 km/hr and crosses a platform in 40 sec. Find the length of the platform.
- (A) 420 m (B) 480 m
(C) 460 m (D) 540 m
- Q6** Train A has 10 compartments and length of each compartment is 30 metres. Train A can cross a bridge of length 260 metres in 28 seconds. With only (Y-2) number of compartments, it takes only 25 seconds to cross the bridge. Find the time taken by train A to cross train B travelling in the opposite direction, if the length of train B is 20Y metres and speed of train B is 25% more than that of train A
- (A) 12 seconds
(B) 10 seconds
(C) $\frac{100}{9}$ seconds
(D) $\frac{25}{2}$ seconds
- Q7** Train A can cross a platform in 10 seconds and a pole in 4 seconds. Train B having $\frac{9}{4}$ times the length of Train A crosses the pole in 7.2 seconds. Find the time taken by train B to cross train A if they are moving in the same direction.
- (A) 9 seconds (B) 13 seconds
(C) 52 seconds (D) 75 seconds
- Q8** Two trains T1 and T2 simultaneously start towards each other at speeds of 40 km/hr and 60 km/hr respectively. An eagle is sitting on the engine of train T1. The moment the trains start, the eagle starts flying towards the second train. Immediately, on reaching the second train, the eagle heads back towards



the first train. Immediately on reaching the first train, the eagle reverses the direction and starts flying towards the second train. This process continues till the two trains meet. If the distance between the trains is 200 km and the speed of the eagle is 80 km/hr, what is the total distance travelled by the eagle in the forward direction (i.e., in the direction towards the second train)?

- (A) 140 km (B) 130 km
(C) 120 km (D) 110 km

Q9 Two trains A and B, travelling in the opposite directions, simultaneously enter a tunnel whose length is equal to the sum of lengths of the two trains. The length of train A is 141m and that of train B is 167m. Both the trains meet after 10 seconds from the point when they were just about to enter the tunnel. If the speeds of the Trains A and B are in the ratio of 4:7, find the length of the train A (in metres) that is outside the tunnel when they meet.

Q10 Indraprastha and Hastinapur are located 60 km apart. Arjun and Bheem begin running from Indraprastha to Hastinapur simultaneously. Arjun, being faster, reaches Hastinapur first and then starts return journey to Indraprastha after taking a rest of 48 minutes. They meet for the first time at a point 12 km away from Hastinapur. Given that there is a 15 km/hr difference in their speeds, what is the total time (excluding the rest time) Arjun takes to complete one whole journey from Indraprastha to Hastinapur and back?

(A) 160 minutes (B) 240 minutes
(C) 320 minutes (D) 400 minutes

Q11 The distance from Point A to Point B is half that of the distance between Point B and Point C. The distance from Point C to Point D is one-quarter of the distance from Point A to Point B. Two trains, Train X and Train Y, travel non-stop from Point A to Point D via Points B and C. The

speed of Train X is twice that of Train Y from Point A to Point B, but the roles reverse when they travel from Point B to Point C. They resume their original speeds while moving from Point C to Point D. What is the ratio of the time taken by Train X compared to Train Y for the entire journey from Point A to Point D?

(A) 7:5 (B) 4:3
(C) 7:6 (D) 8:7

Q12 Buses X and Y operate at constant speeds between Station 1 and Station 2. Bus X departs from Station 1 at 9 am and stops at 9:10 am due to a technical issue. Bus Y leaves Station 2 at 10 am and reaches Bus X at 10:20 am. The crew from Bus Y assists and fixes Bus X's issue by 10:30 am. After the repair, both the buses resume their journey towards their respective destinations, but their speeds are reduced by 60% for Bus X and by one third of its original speed for bus Y due to the incident. Bus Y arrives at Station 1 at 11:00 am. At what time does Bus X reach Station 2?

(A) 10:50 AM (B) 10:55 AM
(C) 11:00 AM (D) 11:10 AM

Q13 Dementus stole an all-terrain vehicle (ATV) parked 66 km away from Furiosa's home and then he started driving it at 55 km/h along a straight path away from her home. Simultaneously, from her home, Furiosa starts chasing Dementus on a bicycle following the same path as Dementus. The speed of the ATV is 1.1 times the speed of the bicycle. If the tailwind gives Furiosa an extra 11 km/h boost in speed after how much time (in minutes) will Furiosa catch up with Dementus? [Neglect the effect of tailwind on the ATV for being massive in size.]

Q14 Mr. Thompson and his assistant, start from their home and check the water level of a reservoir every day at 8 a.m. and then instantly return home together. One day, Mr.



Thompson reached the reservoir an hour earlier i.e. at 7 a.m. and after checking the water level, started walking towards home. His assistant, who started from their home at the same time as usual, was on the way to the reservoir, met Mr. Thompson on the way, took a U-turn and they both returned home 30 minutes earlier than usual. How long (in minutes) did Mr. Thompson walk that day before meeting his assistant? [Provided that no time is lost in the checking process.]

- Q15** The front wheel of a bicycle has a smaller radius than its rear wheel. The radii of the front and rear wheels are 18 cm and 24 cm, respectively. The bicycle covered a distance of 66 km in 1 hour and 40 minutes. Assuming the bicycle maintained a constant speed throughout the journey, find the difference in the number of revolutions per hour between the two wheels.
(A) 7350 (B) 7700
(C) 8400 (D) 8750
- Q16** Every day, I leave home at the same time and arrive at the office right on time. Today, I departed 8 minutes earlier than usual but drove 20% slower than my regular speed. Surprisingly, I reached the office 4 minutes later than the reporting time. Which of the following pairs can be the distance between my home and office and my driving speed today?
(A) 60 km, 75 km/h (B) 50 km, 60 km/h
(C) 40 km, 32 km/h (D) 32 km, 32 km/h
- Q17** Three buses departed simultaneously from Varanasi to Lucknow along the same highway. The second bus traveled 30 km/h faster than the first bus and reached Lucknow two hours earlier than the first bus. The third bus arrived in Lucknow forty minutes after the second bus, covering half the time at the speed of the first bus and the other half at the speed of the

second bus. What is the distance (in km) between Varanasi and Lucknow?

- Q18** A sailboat goes downstream starting with one port onto the next in 2 hours. It covers the same distance upstream in 3 hours. If the speed of the stream is 4 km/h, the distance between the two ports is:
(A) 52 km (B) 48 km
(C) 16 km (D) 24 km
- Q19** The speed of a boat in still water is 20% more than that of the current. After traveling for 4 hours downstream, it traveled for 4 hours upstream, increasing the boat's speed by 15 km/h. The distance covered by the boat downstream is 175% more than that of upstream. Find the original downstream speed of the boat.
(A) 75 km/hr (B) 80 km/hr
(C) 55 km/hr (D) 40 km/hr
- Q20** A student calculated his time to reach school from home while swimming 2.4 km in still water. One day there was a stream of speed 4 km/h and he swam along the stream and reached school 1.2 minutes early. Find the swimming speed (in still water) of the student.
(A) 22 km/h
(B) 24 km/h
(C) 20 km/h
(D) Cannot be determined
- Q21** In still water, the speed of boat A is less than that of boat B, and the difference in their speeds is equal to the speed of the stream. If the upstream speed of boat A is half of the downstream speed of boat B, then find the ratio of the speed of boat A to that of boat B in still water.
(A) 2 : 3 (B) 5 : 6
(C) 4 : 5 (D) 3 : 4
- Q22** Two children Dholu and Bholu start descending on an escalator which is going



down. Bholu is 4 times as fast as Dholu. By the time they reach the bottom, Dholu individually descends 30 steps while Bholu individually descends 40 steps. How many steps are visible on the escalator when it is turned off?

- Q23** In a descending escalator that has 224 steps, Pratap starts running upwards and Bhanu starts running downwards at different speeds at the same time. The speed of the escalator is the difference of thrice the speed of Pratap and six times the speed of Bhanu. Given that the ratio of speeds of Pratap and Bhanu is 5 : 2, find the step of the escalator from the top on which Pratap and Bhanu meet.
- Q24** At 8:00 AM Captain Jack Sparrow started sailing on a ship from point X to Y at a distance of 600 km along the flow of the river. At 10:30 AM after travelling 200 km, his ship's engine was damaged and he immediately left the ship floating and continued his journey on a boat at the same speed as he was sailing initially. He reached Y, then wasted 2 hours on finding a mechanic, and finally sailed back to find his ship. If the speed of the stream is 5 km/h, then when will he find his ship? (Assume that no time is wasted between the damage of the engine and starting his journey on the boat.)
 (A) 10:30 PM (B) 10:22 PM
 (C) 9:30 PM (D) 9:22 PM
- Q25** Two adventurers, Karan and Jamie, are climbing up a moving up escalator that is travelling at a constant speed. Karan takes 4 steps for every 9 steps taken by Jamie. At the same time, Karan reaches the top of the escalator after taking 20 steps, while Jamie takes 36 steps to reach the top. If the escalator were not moving, how many steps would Karan and Jamie need to take to reach the top?
- Q26** A conveyor belt in a factory moves packages at a constant speed. If a worker, Pawan, walks on the conveyor belt in the same direction as it

moves at his normal pace, it takes him 30 seconds to reach the end. If Pawan makes his walking speed thrice, it takes him 50 seconds to reach the end when walking against the conveyor belt's direction. How long would it take (in seconds) for a box of goodies to travel the entire length of the conveyor belt (from one end to another end)?

- Q27** In still water, three boats depart from Dock X to Dock Y at equal time intervals. They reach Dock Y simultaneously and then continue in the same direction and head towards Dock Z, which is 360 km away from Dock Y. The first boat arrives at Dock Z two hours after the second boat. Upon reaching Dock Z, the third boat immediately turns back towards Dock Y. The first and third boats meet at a point 60 km away from Dock Z. What is the difference in the speeds of the first and third boats in km/h, rounded to the nearest integer?
 (A) 7 (B) 8
 (C) 9 (D) 10
- Q28** Bobby takes 40 seconds to ride an escalator when walking at his normal pace in the same direction as the escalator. If he doubles his walking speed, it takes him 80 seconds to ride the escalator in the opposite direction. If Bobby chooses to remain stationary on the escalator, the time required to complete the ride (in seconds) is:
- Q29** A cyclist can ride at 10 km/h. While cycling on a hilly terrain the gravitational pull affects the speed of a cyclist, it has similar effects while going uphill and downhill. The time taken for the cyclist to ride uphill and downhill on a hill between two fixed points is 6 hours and 1.5 hours respectively. If the cyclist's speed is increased by $s\%$, the total time to ride uphill and downhill reduces by 60%. Determine s .
 (A) 80 (B) 150
 (C) 75 (D) 100



Q30 Vanshika and Shivani walk up a moving up escalator at constant speeds. For every 8 steps taken by Vanshika, Shivani takes 3 steps. Given that Vanshika takes 32 and Shivani takes 24

steps to climb up the escalator moving up then how many steps would each have to climb when the escalator is switched off ?



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Answer Key

Q1 8
Q2 13
Q3 B
Q4 4
Q5 D
Q6 C
Q7 C
Q8 C
Q9 29
Q10 B
Q11 C
Q12 B
Q13 660
Q14 45
Q15 D

Q16 D
Q17 120
Q18 B
Q19 C
Q20 C
Q21 C
Q22 45
Q23 160
Q24 B
Q25 100
Q26 50
Q27 D
Q28 80
Q29 A
Q30 40



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Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

Let speed of Priya be x km/h, so

Relative speed of Riya and Priya = $(4+x)$ km/h

Relative speed of Sohan and Priya = $(10+x)$ km/h

So

$$\frac{3}{4+x} - \frac{3}{10+x} = \frac{5}{60}$$

$$\Rightarrow \frac{1}{4+x} - \frac{1}{10+x} = \frac{1}{36}$$

$$\Rightarrow \frac{10+x-4-x}{x^2+14x+40} = \frac{1}{36}$$

$$\Rightarrow x^2 + 14x - 176 = 0$$

$$\Rightarrow x^2 + 22x - 8x - 176 = 0$$

$$\Rightarrow (x+22)(x-8) = 0$$

$$\Rightarrow x = 8 \text{ or } -22$$

Since speed cannot be negative, speed of Priya is 8 km/h.

Video Solution:



Q2 Text Solution:

Relative speed in First minute = $(20+30) = 50$ m/s

Relative speed in second minute = $(18+28) = 46$ m/s

Relative speed in third minute = 42 m/s and so on

So

$338 = 50 + 46 + 42 + \dots + [50 - 4(n-1)]$ (where n is time required in minutes)

$$\Rightarrow 338 = \frac{n}{2} [2 \times 50 + (n-1)(-4)]$$

$$\Rightarrow 676 = n(-4n + 104)$$

$$\Rightarrow n^2 - 26n + 169 = 0$$

$$\Rightarrow (n-13)^2 = 0$$

$$\Rightarrow n = 13$$

They will meet after 13 minutes.

Video Solution:



Q3 Text Solution:

If they would have started simultaneously:

Initial separation between them = 120 km

Relative speed = $24 + 36 = 60$ km/hr

Time taken to meet = $\frac{120}{60} = 2$ hr

Distance travelled by Qutub in 2 hr = $36 \times 2 = 72$ km

So, they would have met 72 km away from point B

In actual case:

Distance travelled by Prateek in 20 minutes =

$$24 \times \frac{20}{60} = 8 \text{ km}$$

Initial separation between them = $120 - 8 = 112$ km

Relative speed = $24 + 36 = 60$ km/hr

Time taken to meet after Qutub starts = $\frac{112}{60} = \frac{28}{15}$ hr

Distance travelled by Qutub in $\frac{28}{15}$ hr = $36 \times \frac{28}{15} = 67.2$ km

So, they met 67.2 km away from B

Required distance between the 2 meeting points = $72 - 67.2 = 4.8$ km

Video Solution:



**Q4 Text Solution:**

The train can cover $(1250 + 950)$ m distance in 40 seconds which means the speed of the train

$$= \frac{2200}{40}$$

$$= 55 \text{ m/s}$$

Relative speed of the dog and train $= 55 - 3 = 52 \text{ m/s}$

To cover the distance of 234 meters, it will take

$$= \frac{208}{52} = 4 \text{ seconds}$$

Video Solution:**Q5 Text Solution:**

The relative speed of train A when it crosses Ranjay

$$= 90 + 18 = 108 \text{ km/hr}$$

$$= 108 \times \frac{5}{18} \text{ m/s}$$

$$= 30 \text{ m/s}$$

$$\text{Length of train A} = 30 \times 24 = 720 \text{ m}$$

$$\text{Length of train B} = 720 + 140 = 860 \text{ m}$$

Let the length of the platform be x m

$$860 + x = 126 \times \frac{5}{18} \times 40$$

$$860 + x = 1400$$

$$x = 540 \text{ m}$$

Video Solution:**Q6 Text Solution:**

$$\text{Length of train A} = 30 \times 10 = 300 \text{ metres}$$

Let the speed of train A be SA m/sec

According to the question,

$$28 = \frac{(260 + 300)}{SA}$$

$$SA = 20 \text{ m/sec}$$

Length of train with $(Y - 2)$ compartments $= 30 \times (Y - 2)$ metres

According to the question,

$$25 = \frac{(260 + 30Y - 60)}{20}$$

$$\Rightarrow 500 = 200 + 30Y$$

$$\Rightarrow Y = 10$$

$$\text{Speed of train B} = \frac{5}{4} \times 20 = 25 \text{ m/sec}$$

$$\text{Length of train B} = 20 \times Y = 200 \text{ metres}$$

$$\text{Required time} = \frac{(300 + 200)}{(20 + 25)} = \frac{100}{9} \text{ seconds}$$

Video Solution:**Q7 Text Solution:**

Let's assume that the speed of the train A is a m/s and that of train B is b m/s.

So, the length of train A is $4a$ meters.

The length of the platforms is $(10 - 4)a$ meters $= 6a$ meters

The length of train B is $9a$ meters

$$\text{Speed of the train B} = 9a / 7.2 \text{ m/s} = 1.25 \text{ m/s}$$

When they are moving in the same direction, the time taken to cross each other will be

$$\frac{4a + 9a}{1.25a - a} \text{ seconds}$$

$$= 52 \text{ seconds.}$$



Video Solution:**Q8 Text Solution:**

The time taken by the two trains to meet

$$= \frac{200}{(40+60)} = 2 \text{ hours}$$

The eagle continuously flies for 2 hours.

Therefore, the distance covered by the eagle

$$= 80 \times 2 = 160 \text{ km.}$$

We can say that, Distance travelled in forward
+ Backward = **160km.**

$$F + B = 160$$

The distance travelled by the first train till the
two trains meet = $40 \times 2 = 80 \text{ km.}$

This means that the eagle finds itself at a point
80 **km** away from the starting point after
having flown **160km.**

$$F - B = 80.$$

On solving both the equations we get,

$$F = 120 \text{ km and } B = 40 \text{ km}$$

Video Solution:**Q9 Text Solution:**

Let the speeds of train A and B be $4v$ and $7v$.

The length of the tunnel = $141 + 167 = 308$ metres.

Since both the trains are travelling in the
opposite directions, relative
velocity = $4v + 7v = 11v$.

(velocity)(time) = tunnel length

$$(11v)(10) = 308$$

$$v = 2.8 \text{ m/s.}$$

Speed of train A is $4(2.8) = 11.2 \text{ m/s.}$

The distance travelled by train A after 10
seconds = $11.2(10) = 112$ metres.

The length of train A that is outside the
tunnel = $141 - 112 = 29$ metres.

Video Solution:**Q10 Text Solution:**

Let's take Arjun's speed as $A \text{ km/h}$ and
Bheem's speed as $B \text{ km/hr.}$

Let's assume Arjun's speed as $x \text{ km/h.}$

Therefore, Bheem's speed will be $(x - 15)$
 km/h.

Time taken by Arjun to reach meeting point:

From Indraprastha to Hastinapur:

$$\text{Time taken} = \frac{60}{x} \text{ hours}$$

From Hastinapur to the meeting point:

$$\text{Time taken} = \frac{12}{x} \text{ hours}$$

So, the total time Arjun takes for the round trip

$$\text{is} = \frac{60}{x} + \frac{12}{x} + \frac{4}{5} = \left(\frac{72}{x} + \frac{4}{5}\right) \text{ hours}$$

Time taken by Bheem to reach meeting point

$$\text{from Indraprastha} = \frac{48}{x-15} \text{ hours}$$

Given that they meet at the same point

$$\frac{72}{x} + \frac{4}{5} = \frac{48}{x-15}$$

$$\Rightarrow 4x^2 + 60x - 5400 = 0$$

Solving we get

$$x = -45, 30$$

Since speed can not be negative, speed of
Arjun is 30 km/h.

So total time taken by Arjun to complete the

$$\text{journey} = \frac{120}{30} = 4 \text{ hours or } 240 \text{ minutes}$$

Video Solution:

**Q11 Text Solution:**

Let the distance between C and D be d km and let the speed of train Y between A and B be y km/h.

	A-B	B-C	C-D
Distance	$4d$	$8d$	d
X's Speed	$2s$	s	$2s$
Y's Speed	s	$2s$	s
X's Time	$\frac{4d}{2s}$ $= 2\left(\frac{d}{s}\right)$	$\frac{8d}{s}$ $= 8\left(\frac{d}{s}\right)$	$\frac{d}{2s}$ $= \frac{1}{2}\left(\frac{d}{s}\right)$
Y's Time	$\frac{4d}{s}$ $= 4\left(\frac{d}{s}\right)$	$\frac{8d}{2s}$ $= 4\left(\frac{d}{s}\right)$	$\frac{d}{s} = \left(\frac{d}{s}\right)$

$$\begin{aligned} \text{Time}_X : \text{Time}_Y &= \left[2\left(\frac{d}{s}\right) + 8\left(\frac{d}{s}\right) + \frac{1}{2}\left(\frac{d}{s}\right) \right] : \left[4\left(\frac{d}{s}\right) + 4\left(\frac{d}{s}\right) + \left(\frac{d}{s}\right) \right] = \frac{21}{2}\left(\frac{d}{s}\right) : 9\left(\frac{d}{s}\right) = 7 : 6 \end{aligned}$$

Video Solution:**Q12 Text Solution:**

Let the speeds of the buses X and Y be x km/min and y km/min.

Distance travelled by X before stopping = $10x$ km

Distance travelled by Y to meet X = $20y$ km

New speed of bus X = $0.4x$ km/min

New speed of bus Y = $\frac{2}{3}y$ km/min

So distance travelled by Y after the repair = $30\left(\frac{2}{3}y\right) = 20y$ km

So

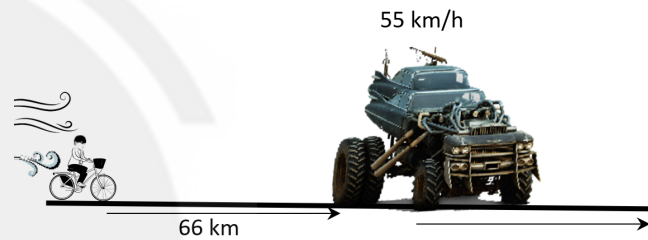
$$10x = 20y$$

$$\Rightarrow x = 2y$$

So the time required by bus X to cover the meaning distance at reduced speed =

$$\frac{20y}{0.4x} = \frac{20y}{0.8y} = 25 \text{ min}$$

So the bus X will reach station 2 at 10:55 AM.

Video Solution:**Q13 Text Solution:**

Dementus got a head-start of 66 km by distance.

The speed of Dementus on ATV is 55 km/h.

So, the actual speed of the bicycle of Furiosa =

$$\frac{55}{1.1} = 50 \text{ km/h}$$

Due to the tailwind, Furiosa's bicycle got an extra boost in speed of 11 km/h.

So, the speed of Furiosa = $50 + 11 = 61$ km/h

Therefore, the relative speed between Furiosa and Dementus = $61 - 55 = 6$ km/h

Now, the distance between them is 66 km. So, if Furiosa covers this distance, she will catch Dementus.

$$\text{So, the required time to catch him} = \frac{66}{6} = 11 \text{ hours} = 660 \text{ minutes.}$$

Video Solution:

**Q14 Text Solution:**

We see that his assistant reduces his total travel time by 30 minutes.

So, his assistant reduces its **one-way**

travelling time by $\frac{30}{2} = 15$ minutes.

Therefore, the assistant would have met Mr. Thompson 15 minutes before 8:00 a.m. i.e. 7:45 a.m.

This means that Mr. Thompson walked from 7 a.m. to 7.45 a.m., i.e., 45 minutes.

Video Solution:**Q15 Text Solution:**

Difference in the number of revolutions per minute between the two wheels

$$= \frac{\frac{66 \times 1000 \times 100}{2 \times \frac{22}{7} \times 18} - \frac{66 \times 1000 \times 100}{2 \times \frac{22}{7} \times 24}}{\frac{5}{3}}$$

$$= \frac{66 \times 1000 \times 100 \times 3 \times 7}{2 \times 22 \times 5} \left(\frac{1}{18} - \frac{1}{24} \right)$$

$$= 3 \times 100 \times 100 \times 3 \times 7 \times \left(\frac{1}{72} \right)$$

$$= 8750$$

Video Solution:**Q16 Text Solution:**

Let the usual speed be x km/hr and the distance between the office and home be y km.

Given, at 20% slower speed ($\frac{4}{5}x$), it takes 12 minutes ($\frac{1}{5}h$) more to cover the distance of y km, so

$$\frac{y}{x} + \frac{1}{5} = \frac{y}{\frac{4}{5}x}$$

$$\Rightarrow \frac{5y+x}{5x} = \frac{5y}{4x}$$

$$\Rightarrow 20y + 4x = 25y$$

$$\Rightarrow x = \frac{5}{4}y$$

So

$$\frac{4}{5}x = \frac{4}{5} \times \frac{5}{4}y = y$$

So the reduced speed and distance between the office and home is same.

Video Solution:**Q17 Text Solution:**

Let the speed of the first bus be x kmph and time required for it to travel be t , so speed of second bus will become $(x+30)$ kmph and time taken by it will be $(t-2)$ hours.

So

$$xt = (x + 30)(t - 2)$$

$$\Rightarrow xt = xt + 30t - 2x - 60$$

$$\Rightarrow t = \frac{x+30}{15}$$

For the third bus,



$$xt = x\left(\frac{t}{2} - \frac{2}{3}\right) + (x + 30)\left(\frac{t}{2} - \frac{2}{3}\right)$$

$$\Rightarrow xt = (2x + 30)\left(\frac{3t-4}{6}\right)$$

$$\Rightarrow 6xt = 6xt - 8x + 90t - 120$$

$$\Rightarrow 90t = 8x + 120$$

$$\Rightarrow 6(x + 30) = 8x + 120$$

$$\Rightarrow 6x + 180 = 8x + 120$$

$$\Rightarrow x = 30 \text{ kmph}$$

So

$$t = \frac{30+30}{15} = 4 \text{ hours}$$

So distance = $xt = 120 \text{ km}$

Video Solution:



Q18 Text Solution:

Let the distance from one port to another is = d

Let the speed of the sailboat is = x

Given, time downstream = 2 hr

$$\text{Time} = \frac{\text{distance}}{(\text{Speed in downstream})}$$

Speed of sailboat downstream = speed of boat + speed of the stream

speed of sailboat in downstream = $x + 4$

$$2 = \frac{d}{(x + 4)}$$

$$d = 2x + 8$$

Given, time in upstream = 3 hr

$$\text{Time} = \frac{\text{distance}}{(\text{Speed in upstream})}$$

Speed of sailboat upstream = speed of the boat - speed of the stream

speed of sailboat upstream = $x - 4$

$$3 = \frac{d}{(x - 4)}$$

$$d = 3x - 12$$

Distance will be equal so

$$2x + 8 = 3x - 12$$

$$x = 20$$

Distance = speed $\times$ time

$$d = (20 - 4) \times 3$$

$$d = 48 \text{ km}$$

Video Solution:



Q19 Text Solution:

Let the speed of the current be 'x' km/hr,

Therefore, the speed of the boat in still water = $1.2x \text{ km/hr}$

Downstream distance traveled by boat = $(x + 1.2x) \times 4 = 8.8x \text{ km}$

Upstream distance covered by the boat = $(1.2x + 15 - x) \times 4 = (0.8x + 60) \text{ km}$

According to the question,

$$\left\{ \frac{8.8x - 0.8x - 60}{0.8x + 60} \right\} = 1.75$$

Or, $x = 25 \text{ km/hr}$

Therefore, original downstream speed of the boat = $2.2x = 55 \text{ km/hr}$

Video Solution:



Q20 Text Solution:

Let the swimming speed in still water = $x \text{ km/h}$

Let calculated time = ' t ' min

According to the question,

$$t = \frac{2.4}{x} \times 60 \dots\dots (1)$$

$$(t - 1.2) = \frac{2.4}{x + 4} \times 60 \dots\dots (2)$$

From (1) and (2), we have

$$\left(\frac{2.4}{x} \times 60 - 1.2 \right) = \frac{2.4}{x + 4} \times 60$$

$$\frac{120}{x} - 1 = \frac{120}{x + 4}$$

$$x = 20, -24$$

But the speed cannot be negative.



Thus, the swimming speed in still water = 20 km/h

Video Solution:



Q21 Text Solution:

Let the speed of boats A and B in still water are 'a' and 'b' respectively.

So, the speed of the stream = (b - a)

Upstream speed of boat A = a - (b - a) = 2a - b

Downstream speed of boat B = b + (b - a) = 2b - a

From the question:

$$2a - b = \frac{1}{2} * (2b - a)$$

$$4a - 2b = 2b - a$$

$$5a = 4b$$

$$\text{Required ratio} = a : b = 4 : 5$$

Video Solution:



Q22 Text Solution:

Let the escalator takes m steps when Dholu descends 30 steps.

Since Bholu is 4 times as fast as Dholu, Dholu would have taken $\frac{40}{4} = 10$ steps by the time Bholu takes 40 steps.

The escalator would have taken $\frac{m}{3}$ steps when Dholu descends 10 steps because it takes m steps when Dholu takes 30.

$$30 + m = 40 + \frac{m}{3}$$

$$m = 15$$

$$\text{Steps Visible} = 30 + m = 30 + 15 = 45$$

Video Solution:



Q23 Text Solution:

Let Pratap's speed and Bhanu's speed be $5x$ and $2x$ respectively.

$$\Rightarrow \text{Speed of escalator} = 15x - 12x = 3x.$$

Pratap's effective speed upwards

$$= 5x - 3x = 2x$$

Bhanu's effective speed downwards

$$= 2x + 3x = 5x$$

Hence, the distance covered by them when they meet will be in the ratio of their speeds i.e., 2:5

Let Pratap cover $2a$ steps, hence Bhanu will have covered $5a$ steps.

$$\text{Total steps} = 2a + 5a = 224.$$

$$\text{Hence, } 7a = 224$$

$$a = 32$$

Hence, the two meet at the 64th step from bottom or $224 - 64 = 160$ th step from the top.

Video Solution:



Q24 Text Solution:

The distance between X and Y = 600 KM

Jack Sparrow travelled 200 KM distance within 2.5 hours (from 8:00 AM to 10:30 AM).

So, the speed of Jack Sparrow (i.e., the speed of his ship in downstream) = $\frac{200}{2.5} = 80$ km/h

The speed of the stream = 5 km/h



So, the actual speed of the ship = $80 - 5 = 75$ km/h

Now, since, the speed of his boat was the same as he was sailing initially.

So, the speed of his boat in downstream = 80 km/h

i.e., the time taken to reach the rest of the distance, i.e., $(600 - 200) = 400$ km is = $\frac{400}{80} = 5$ hours

Thus, he will reach point Y at $(10:30 \text{ AM} + 5 \text{ hours}) = 3:30 \text{ PM}$.

Then, he wasted another 2 hours on finding a mechanic.

Finally, he started sailing back to find his ship at $(3:30 \text{ PM} + 2 \text{ hours}) = 5:30 \text{ PM}$.

Now, in these $(5 + 2) = 7$ hours, the engine-damaged ship will not stay stationary. It will also move along the flow of the river at 5 km/h.

So, in 7 hours, it moved = $(7 \times 5) = 35$ km towards the point Y.

i.e., now, Captain Jack Sparrow has to cover only $(400 - 35) = 365$ km to find his ship.

At this point, the speed of Jack Sparrow in his boat against the flow of the river = $(75 - 5) = 70$ km/h

So, the relative speed between his boat and his ship = $(70 + 5) = 75$ km/h

Hence, Jack Sparrow will find his ship in =

$$\frac{365}{75} = \frac{73}{15} = 4\frac{13}{15} \text{ hours}$$

=

$$4 \text{ hours} + \frac{13}{15} \times 60 \text{ minutes} = 4 \text{ hours } 52 \text{ minutes}$$

i.e., He will find his ship at $(5:30 \text{ PM} + 4 \text{ hours } 52 \text{ minutes}) = 10:22 \text{ PM}$

Video Solution:



Q25 Text Solution:

Let the speed be measured by how many steps are taken per second.

Karan takes 4 steps = Jamie takes 9 steps.

i.e., the ratio of speed, Jamie : Karan = 4 : 9
When the escalator is not moving, let x steps are visible.

Given that, Karan takes 20 steps to reach the top while Jamie takes 36.

When moving up, the ratio of speed of Karan to the escalator = $\frac{20}{x-20}$ [Note that, from x visible steps, Karan takes 20 while Escalator takes the rest, i.e., $(x - 20)$ steps.]

Similarly, the ratio of speed of Jamie to the escalator = $\frac{36}{x-36}$

So, the ratio of speed, Jamie : Karan =

$$\frac{\text{Speed of Karan}}{\text{Speed of Escalator}} = \frac{\text{Speed of Escalator}}{\text{Speed of Jamie}} = \frac{4}{9}$$

$$\frac{20}{x-20} = \frac{4}{9}$$

$$\frac{x-20}{36} = \frac{4}{9}$$

$$9 \times \frac{20}{x-20} = 4 \times \frac{36}{x-36}$$

$$\frac{5}{x-20} = \frac{4}{x-36}$$

$$x = 180 - 80 = 100$$

Therefore, both of them need to take 100 steps to reach the top.

Video Solution:



Q26 Text Solution:



Let the normal walking speed of Pawan = x steps/second

and the speed of the conveyor belt = y steps/second

The time taken by Pawan to ride the entire conveyor belt at his normal speed along the moving direction of the conveyor belt is 30 seconds.

Total Distance = $(x + y) \times 30$ steps

When he made thrice his walking speed to ride the conveyor belt, it took 50 seconds to ride the conveyor belt in the opposite direction.

Total Distance = $(3x - y) \times 50$ steps

Therefore, we have

$$(x + y) \times 30 = (3x - y) \times 50$$

$$120x = 80y$$

$$3x = 2y$$

Hence, for a box to travel the entire length of the conveyor belt, it takes

$$= \frac{(x+y) \times 30}{y} = \frac{(3x+3y) \times 10}{y} = \frac{5y \times 10}{y} = 50$$

seconds, since a box is an object left stationary, so the travel speed is same as the speed of the conveyor belt.

Video Solution:



Q27 Text Solution:

Since, the first and the third boats meet after travelling for equal intervals of time, so the ratio of the speed of the first boat to that of the third boat = $(360 - 60) : (360 + 60) = 5 : 7$... (i)

Let the distance from Dock X to Y = a km

And the interval between the departure of the first boat and the second boat = t

Total time taken by first boat to cover the distance from Dock X to Y = T

So, the speed of the first boat = $\frac{a}{T}$ km/h

The speed of the second boat = $\frac{a}{T-t}$ km/h

The speed of the third boat = $\frac{a}{T-2t}$ km/h

[Since, each of them departs at equal time intervals.]

Therefore, we have

$$\frac{\frac{a}{T}}{\frac{a}{T-2t}} = \frac{5}{7}$$

$$\frac{T-2t}{T} = \frac{5}{7}$$

$$2T = 14t$$

$$T = 7t \dots (ii)$$

Now, given that, from the Dock Y, the first boat arrives at Dock Z an hour after the second boat.

$$\text{So, } \frac{360}{\frac{a}{T}} - \frac{360}{\frac{a}{T-t}} = 2$$

$$\frac{360T}{a} - \frac{360(T-t)}{a} = 2$$

$$\frac{T}{a} - \frac{T-t}{a} = \frac{2}{360} \quad \left[\text{by } (ii) \right]$$

$$T - \frac{6T}{7} = \frac{a}{180}$$

$$\frac{T}{7} = \frac{a}{180}$$

$$\frac{a}{T} = \frac{180}{7} \text{ km/h}$$

So, the speed of the first boat = $\frac{180}{7}$ km/h

Therefore, the speed of the third boat =

$$\frac{7}{5} \times \frac{180}{7} = 36 \text{ km/h [By (i)]}$$

Hence, the required speed difference between the first and third boat =

$$36 - \frac{180}{7} = \frac{252-180}{7} = \frac{72}{7} \approx 10 \text{ km/h}$$

Video Solution:



Q28 Text Solution:



Let the normal walking speed of Bobby = x steps/ second

and the speed of the escalator = y steps/second

The time taken by Bobby to ride the entire escalator with his normal speed along the moving direction of the escalator is 40 seconds.

Total Distance = $(x + y) \times 40$ steps

When he doubles his walking speed to ride the escalator, it takes 80 seconds to ride the escalator in the opposite direction.

Total Distance = $(2x - y) \times 80$ steps

Therefore, we have

$$(x + y) \times 40 = (2x - y) \times 80$$

$$120x = 120y$$

$$x = y$$

Now, when Bobby chooses to travel the same escalator in stationary mode, then he will take,

$$\text{time} = \frac{(x+y) \times 40}{y} = \frac{(y+y) \times 40}{y} = 80 \text{ seconds}$$

Video Solution:



Q29 Text Solution:

Let the speed of the cyclist be c and the effect due to gravitation pull on cyclist be g km/h.

Also let the distance between the two points be d km.

So $c = 10$ km/h

Time taken to travel uphill = 6

$$\Rightarrow \frac{d}{10-g} = 6$$

$$\Rightarrow d = 6(10 - g)$$

Time taken to travel downhill = 1.5

$$\Rightarrow \frac{d}{10+g} = 1.5$$

$$\Rightarrow 6(10 - g) = 15 + 1.5g$$

$$\Rightarrow 60 - 6g = 15 + 1.5g$$

$$\Rightarrow g = 6 \text{ km/h}$$

So $d = 24$ km

Total travel time normally = $6 + 1.5 = 7.5$ hours

So new reduced total travel time = $7.5 \times 0.40 = 3$ hours

Let's consider the new speed of cyclist be a km/h, so

$$\frac{24}{a+6} + \frac{24}{a-6} = 3$$

$$\Rightarrow 24 \left(\frac{1}{a+6} + \frac{1}{a-6} \right) = 3$$

$$\Rightarrow 8 \left(\frac{a-6+a+6}{a^2-36} \right) = 1$$

$$\Rightarrow \frac{16a}{a^2-36} = 1$$

$$\Rightarrow a^2 - 16a - 36 = 0$$

$$\Rightarrow (a - 18)(a + 2) = 0$$

$$\Rightarrow a = 18 \text{ km/h}$$

So the new speed of cyclist is 18 km/h

So

$$10 \left(1 + \frac{s}{100} \right) = 18$$

$$\Rightarrow 1 + \frac{s}{100} = \frac{9}{5}$$

$$\Rightarrow \frac{s}{100} = \frac{4}{5}$$

$$\Rightarrow s = 80\%$$

Video Solution:



Q30 Text Solution:

Let the speed of the escalator be 'e' steps per second and Vanshika's speed be 8 steps/sec and Shivani's be 3 steps/sec.

Relative speed is the speed of (person + stairs).

Vanshika walks 32 steps in 4 seconds so in 4 seconds, the escalator will move 4e steps.



Hence total distance covered = $32 + 4e$

Similarly, Shivani walks 24 steps in 8 seconds so in 8 seconds, the escalator will move $8e$ steps.

Total distance covered is same in both the cases

$$32 + 4e = 24 + 8e$$

Therefore, $e = 2$

Hence, the total length of the escalator is $32 +$

$$4 \times 2 = 40 \text{ steps.}$$

Video Solution:



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[iOS App](#)



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